N83: full-size prebuilt U/D public-query panel

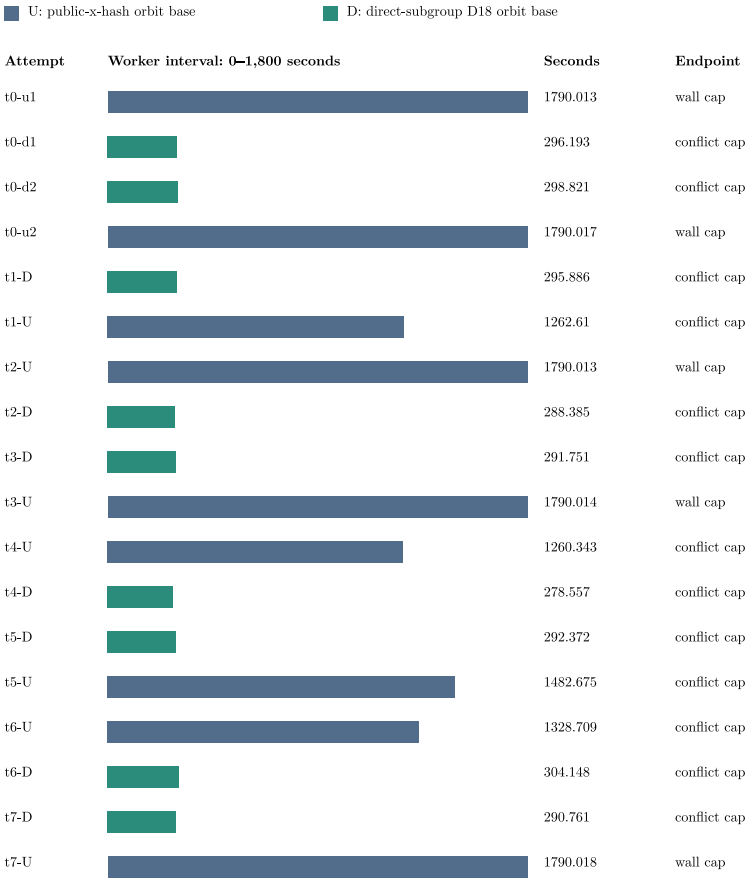
Frozen bounded m=4 comparison · 2026-10-20 UTC

The card started at 2026-10-20T02:06:17Z and ended at 2026-10-20T06:48:53Z. The native receipt audit accepts 18 natural attempts, four constructed controls and four separate general-field control replays. Its complete inventory contains 329 files.

N83 m=4: full-size U/D public searches

16,627 columns / 2,760,082 points per base · eight common public points

Each bar is one retained attempt, in execution order. Shared host: L0.



Limits: 10,000 conflicts, 1,790 s internal / 1,800 s outer wall. The endpoints differ across attempts. These bars do not give a time ratio for equal work or a cost per relation. Every cap remains censored. Repeated target-0 attempts are displayed individually; other points have one attempt per base.

Latest durable completed-conflict counts

Attempt	Conflicts	Decisions	Unobserved tail (s)
t0-u1	2877	827319	3.986
t0-d1	10000	51772	0.01
t0-d2	10000	51772	0.012
t0-u2	2822	816126	4.12
t1-D	10000	51770	0.012
t1-U	10000	512363	0.015
t2-U	4196	773723	4.174
t2-D	10000	51798	0.013
t3-D	10000	51794	0.012
t3-U	4195	765714	3.871
t4-U	10000	512363	0.018
t4-D	10000	51787	0.012
t5-D	10000	51795	0.015
t5-U	10000	585753	0.033
t6-U	10000	512363	0.014
t6-D	10000	51804	0.013
t7-D	10000	51789	0.016
t7-U	4194	739312	3.991

Wall-cap counters come from the last durable progress sample and are lower bounds on completed work. Final conflict-cap counters are bound to the returned receipt. Source: full-m4-v1/final/audit.json and its content-addressed archive. Worker revision d9ef42301cc3; protocol full-m4-common-public/v1.

Exact inputs and policy

Curve `icv1-f2m83-tm6151469093347-debefd74`, a=0, subgroup 241785163923079621668568 9, cofactor 4, polynomial z^83+z^45+z^2+z+1. U is the retained public-x-hash orbit base; D is the exported direct-subgroup dimension-18 base. Each contains 16,627 signed-Frobenius columns and 2,760,082 points. Their accepted point sets differ.

Both bases use the same eight public points, worker binary, solver policy, resource limits and deterministic search behavior. Validation scalars are excluded. Target 0 uses U,D,D,U; odd later indices use D,U and even indices use U,D. There is one model per attempt, four free summands, free selectors and phases, 10,000 conflicts, 300,000 variables and 8,000,000 domain clauses. Each actual model has 201,598 variables, 5,915,238 clauses and 3,901 XOR rows. The satisfied-row cache is enabled for every natural attempt.

The four constructed controls, cache off/on within each base, passed exact SAT-model, row and search-work checks before public search. Their separate general-field replays reject changed targets, points and coefficients. Those constructed rows are excluded from the natural rank calculation.

Observed outcomes and charges

U: 9 attempts; 14284.412 seconds summed over all worker/cap intervals. Outcomes: (UNKNOWN_budget: 5, UNKNOWN_search_cap: 4). Accepted independently replayed natural rows: 0 retained factor-row rank: 0.

D: 9 attempts; 2636.873 seconds summed over all worker/cap intervals. Outcomes: (UNKNOWN_search_cap: 9). Accepted independently replayed natural rows: 0 retained factor-row rank: 0.

The rank is the rank of the retained accepted factor rows modulo the exact subgroup order, separately for U and D. It excludes dependent rows. It does not infer natural yield or a complete IC rank from censored searches. The natural-yield estimate, useful-rank panel, candidate and best total runtime remain null. Different cap endpoints do not support a U/D speed ratio.

Worker intervals include import/index, corpus validation, model construction, root propagation and search up to the receipt timestamp or internal cap. They end before final receipt serialization. Container start/end receipts preserve the surrounding intervals. Construction/export, full-base verification, control/replay processes, Docker and custody costs remain separately recorded; they are not silently included in the worker sum. Latest durable CPU/RSS observations on wall caps are lower bounds. A final cgroup peak is absent for those cap receipts. Missing quantities remain missing.

Host, custody and audit

Intel Xeon 6975P-C SSH Spot runner, pinned CPU 31, one CPU quota, hard 4 GiB, zero swap, no worker network and read-only inputs/root. Other workloads share the host, so isolation is L0. No new instance or remote compilation was introduced during this card.

Worker source: d9ef42301cc3d22f437e1f89304e673836afb6a3. Archive SHA-256: b2921b2e1c2cbb7c8b2030b1dc4e24acfee781546a3691bbe76d24ef9f960c78. Manifest SHA-256: 570b6a102741eae862a638dcd9b0bfd08e0f8d8e6fb28a3a2904f53e2e3218a9.

`examples/n83_full_m4_audit.rs` checks the full inventory; pinned source, binary, corpus and object identities; exact arguments and input mounts; container image/resources, exits and serial order; phase/progress records; counter and resource monotonicity; final/cap consistency; independent replay bindings; and retained factor-row rank. A wall cap cannot be promoted to a completed conflict endpoint or a mathematical UNSAT result.

Original gates and next obligation

The D18 full object and generic replay are complete. Durable m=4 progress and this frozen equal-size public panel are retained. The same-set prebuilt/on-demand m=2 and m=3 finite-policy measurements remain a separate experiment. Complete IC useful-rank collection, linear algebra, individual logarithm verification, qualified cold/reuse cost, matched same-point rho, external-fastest comparison and best-base selection remain open.

The next discriminating stage must obtain independently checked natural rank increments under a separately frozen search or partial-relation policy. The existing caps remain in the record. This panel supplies bounded search evidence, with no qualified pipeline or fastest-base promotion.

Sources: `FULL_M4_PROTOCOL.md`, `FULL_M4_HOST_CARD.md`, `full-m4-v1/final/`, this report's `RUN_REPORT.md`, and `problems/n83-prebuilt-base-quality-m2-m4.md`. Earlier interrupted and smaller-base cards remain separate. The SVG and 150 dpi PNG share this plotting source; matching report and visual copies are in `benchmarks/`.